Web Appendix to "How Puzzling Is the Forward Premium Puzzle? A Meta-Analysis"*

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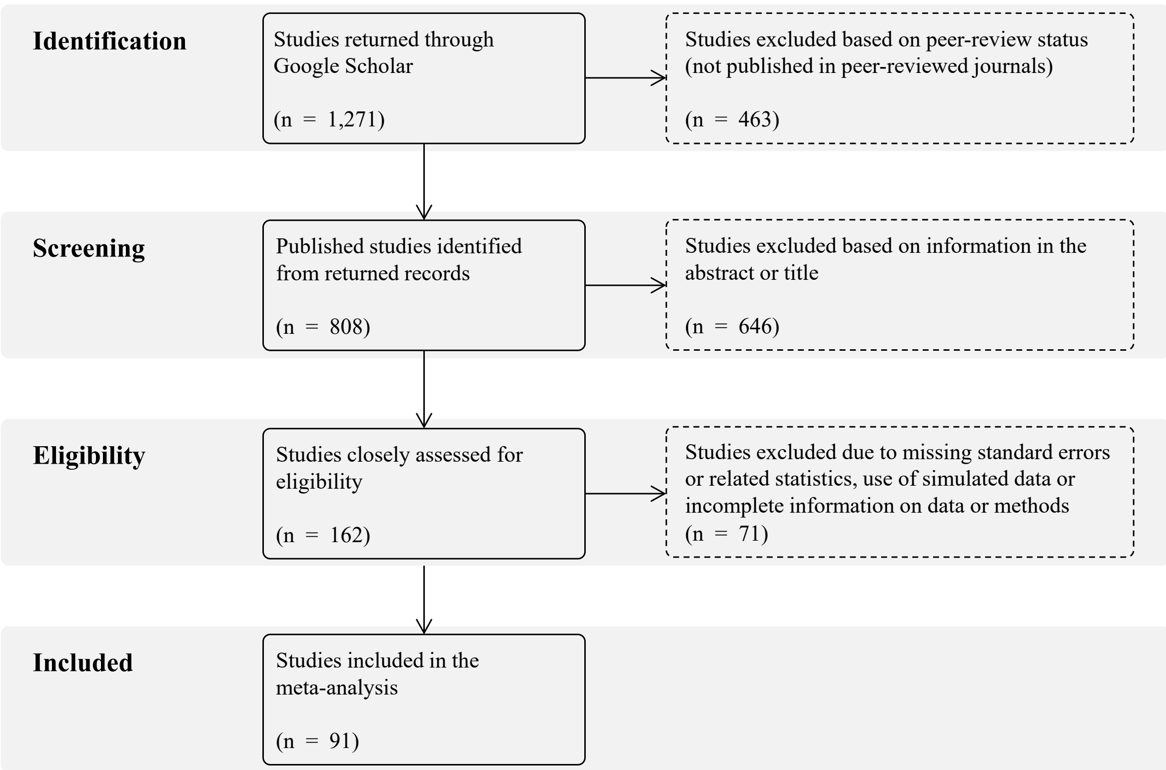
Abstract

This appendix provides details on the selection of studies, implementation of BMA, robustness checks, and a list of studies included in the meta-analysis.

1 Details on the Selection of Studies

Figure 1: *A PRISMA flow diagram*



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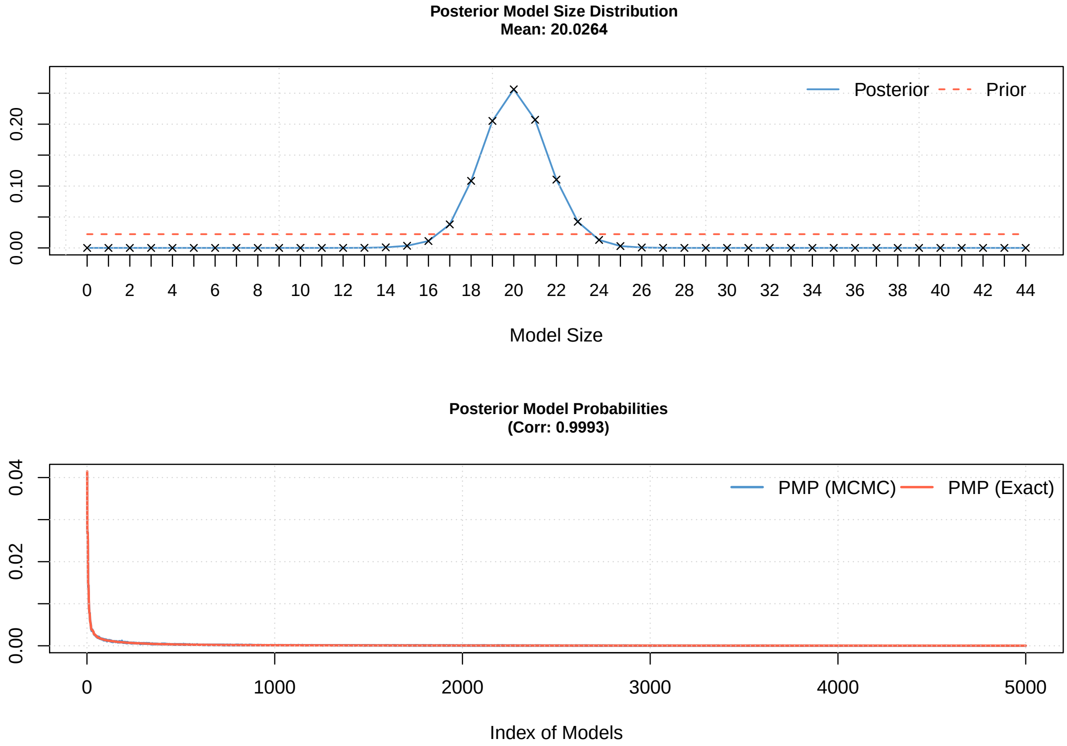
2 BMA Diagnostics and Robustness Checks

Table 1: *Diagnostics of the baseline BMA estimation*

Mean no. regressors	*Draws*	*Burn-ins*	*Time*	*No. models visited*
20.0264	$5 \cdot 10^6$	$1 \cdot 10^6$	7.664059 mins	751,865
Modelspace	*Visited*	*Topmodels*	*Corr PMP*	*No. obs.*
$1.8 \cdot 10^{13}$	0.0000043%	78%	0.9993	2,989
Model prior	*g-prior*	*Shrinkage-stats*		
Random-dilution / 22	UIP	Av = 0.9997		

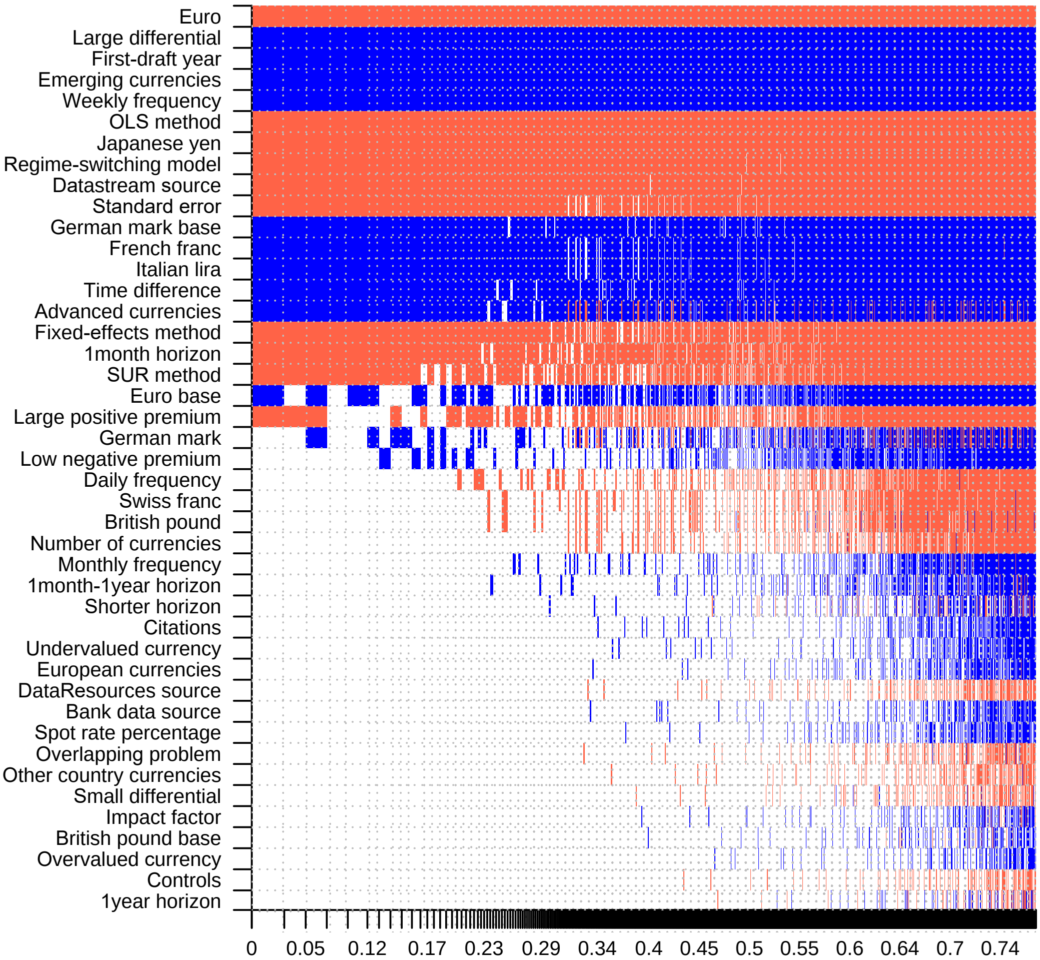
Notes: In the baseline model, we employ the priors suggested by Eicher *et al.* (2011), who recommend using the unit information prior (the prior has the same weight as one observation of data) and using the dilution prior suggested by George (2010), which accounts for collinearity.

Figure 2: *Model size and convergence of the baseline BMA estimation*



Notes: The figure depicts the posterior model size distribution and the posterior model probabilities of the BMA exercise.

Figure 3: *Model inclusion in BMA (BRIC and random priors)*



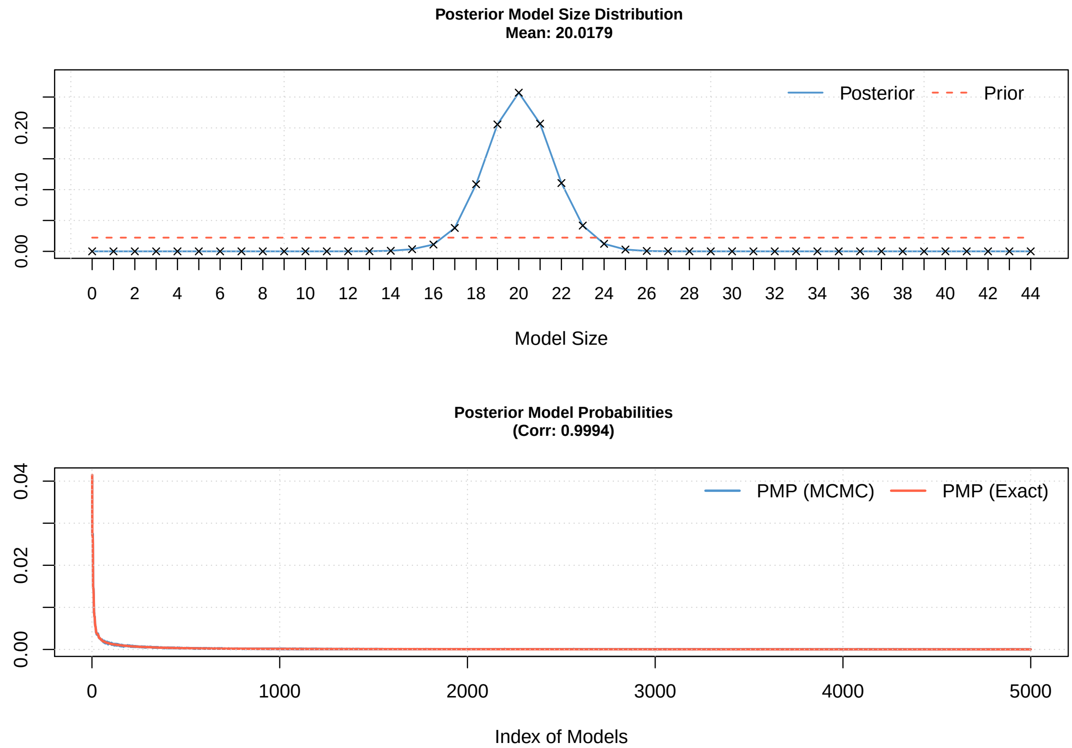
Notes: The figure depicts results of the BMA model, where the response variable is an estimate of slope coefficient *beta* from the difference regressions. In this sepcification, we employ BRIC g-prior suggested by Fernandez *et al.* (2001) and the beta-binomial (random) model prior according to Ley & Steel (2009). Columns show individual models, and variables are listed in descending order by their posterior inclusion probabilities. The horizontal axis shows cumulative posterior model probabilities from the 5,000 best models. Blue color (darker in grayscale) = the variable is included in the model with a positive sign. Red color (lighter in grayscale) = the variable is included in the model with a negative sign. No color = the variable is missing from the model.

Table 2: *Diagnostics of the BMA estimation (BRIC and random priors)*

Mean no. regressors	*Draws*	*Burn-ins*	*Time*	*No. models visited*
20.0179	$5 \cdot 10^6$	$1 \cdot 10^6$	8.213935 mins	750,145
Modelspace	*Visited*	*Topmodels*	*Corr PMP*	*No. obs.*
$1.8 \cdot 10^{13}$	0.0000043%	78%	0.9994	2,989
Model prior	*g-prior*	*Shrinkage-stats*		
Random / 22	BRIC	Av = 0.9997		

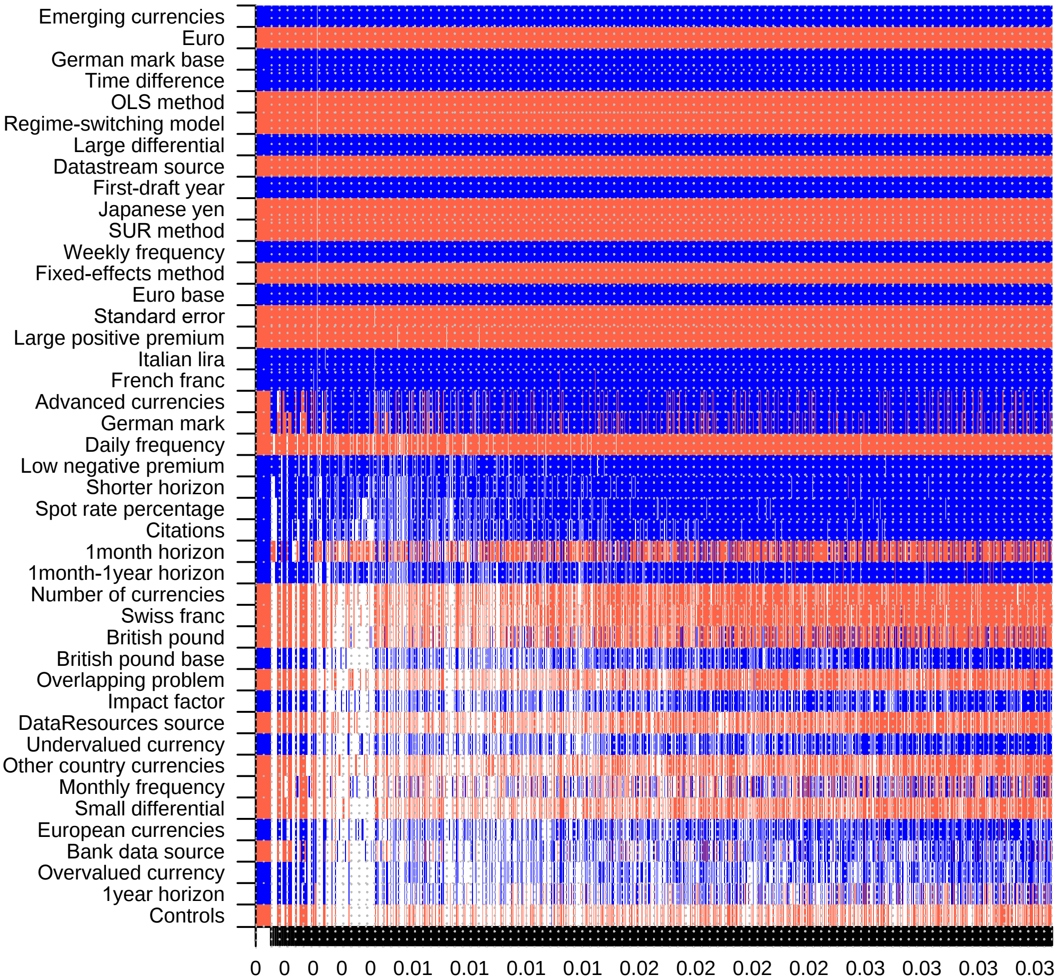
Notes: The specification uses a BRIC g-prior suggested by Fernandez *et al.* (2001) and the beta-binomial model prior according to Ley & Steel (2009).

Figure 4: *Model size and convergence of the BMA estimation (BRIC and random priors)*



Notes: The figure depicts the posterior model size distribution and the posterior model probabilities of the BMA exercise using the BRIC prior for estimated parameters and the random prior for estimated models.

Figure 5: *Model inclusion in BMA (hyper-g and random priors)*



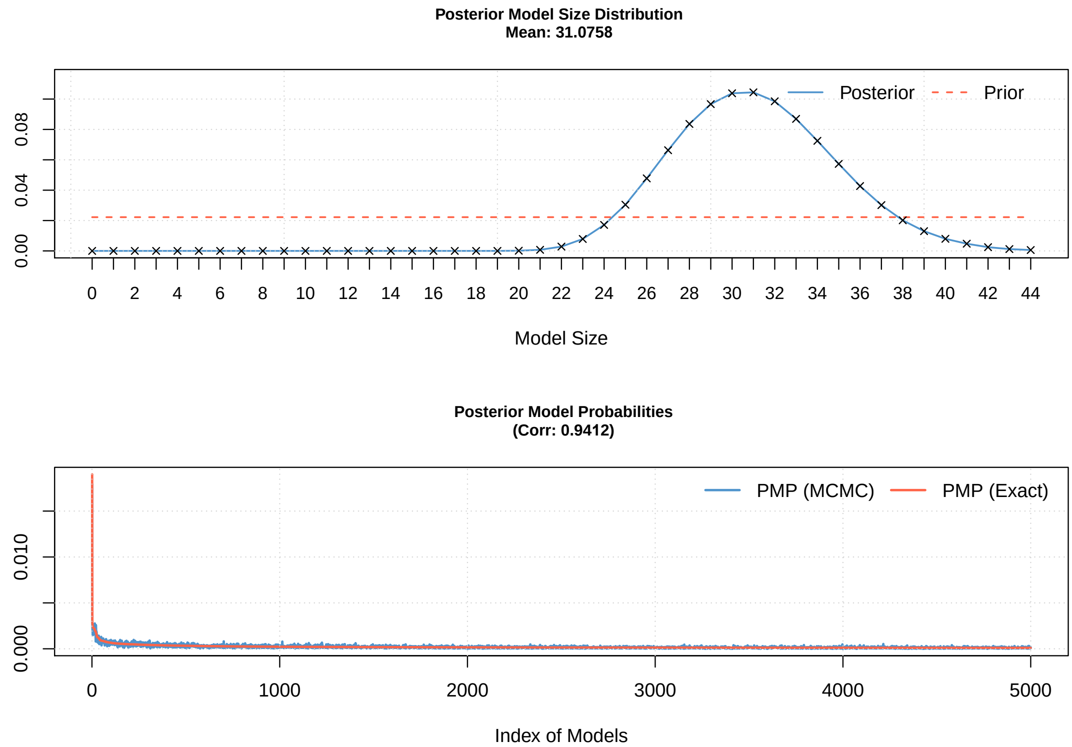
Notes: The figure depicts results of the BMA model, where the response variable is an estimate of slope coefficient *beta* from the differences regressions. In this specification, we employ the data-dependent hyper-g prior suggested by Feldkircher & Zeugner (2012) and the beta-binomial (random) model prior according to Ley & Steel (2009). Columns show individual models, and variables are listed in descending order by their posterior inclusion probabilities. The horizontal axis shows cumulative posterior model probabilities from the 5,000 best models. To ensure convergence of the Markov Chain Monte Carlo sampler, we use 5,000,000 iterations with 1,000,000 burn-ins to allow the sampler to converge to the part of the model space with high posterior probability models. Blue color (darker in grayscale) = the variable is included in the model with a positive sign. Red color (lighter in grayscale) = the variable is included in the model with a negative sign. No color = the variable is missing from the model.

Table 3: *Diagnostics of the BMA estimation (hyper-g and random priors)*

Mean no. regressors	*Draws*	*Burn-ins*	*Time*	*No. models visited*
31.0758	$5 \cdot 10^6$	$1 \cdot 10^6$	20.05682 mins	2,118,785
Modelspace	*Visited*	*Topmodels*	*Corr PMP*	*No. obs.*
$1.8 \cdot 10^{13}$	0.000012%	3.3%	0.9412	2,989
Model prior	*g-prior*	*Shrinkage-stats*		
Random / 22	hyper (a=2.001)	Av = 0.9746		
		St. dev. = 0.007		

Notes: The specification reported in Figure 5 uses the data-dependent hyper-g prior suggested by Feldkircher & Zeugner (2012) and a random model prior advocated by Ley & Steel (2009).

Figure 6: *Model size and convergence of the BMA estimation (hyper-g and random priors)*



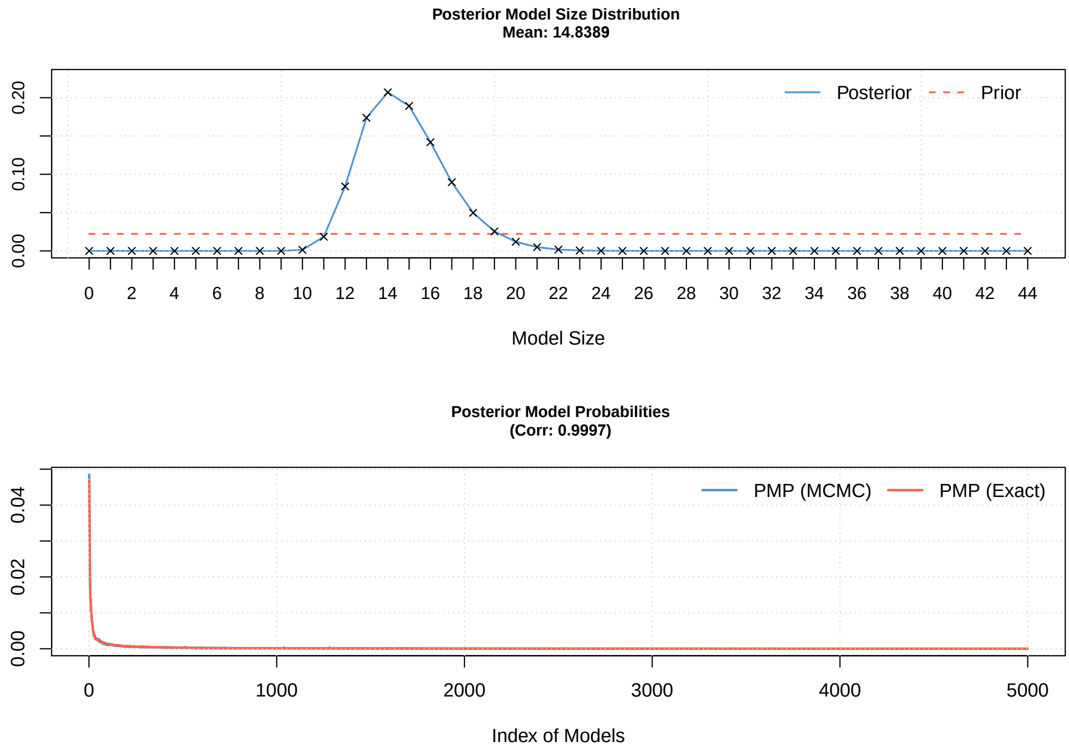
Notes: The figure depicts the posterior model size distribution and the posterior model probabilities of the BMA exercise using the hyper-g prior for estimated parameters and random prior for estimated models (Figure 5).

Table 4: *Diagnostics of the BMA estimation (trimmed data)*

Mean no. regressors	*Draws*	*Burn-ins*	*Time*	*No. models visited*
14.8389	$5 \cdot 10^6$	$1 \cdot 10^6$	9.700674 mins	694,939
Modelspace	*Visited*	*Topmodels*	*Corr PMP*	*No. obs.*
$1.8 \cdot 10^{13}$	0.000004%	79%	0.9997	2,989
Model prior	*g-prior*	*Shrinkage-stats*		
Random-dilution / 22	UIP	Av = 0.9996		

Notes: In the trimming-data model, we employ the priors suggested by Eicher *et al.* (2011), who recommend using the unit information prior (the prior has the same weight as one observation of data) and using the dilution prior suggested by George (2010), which accounts for collinearity.

Figure 7: *Model size and convergence of the BMA estimation (trimmed data)*



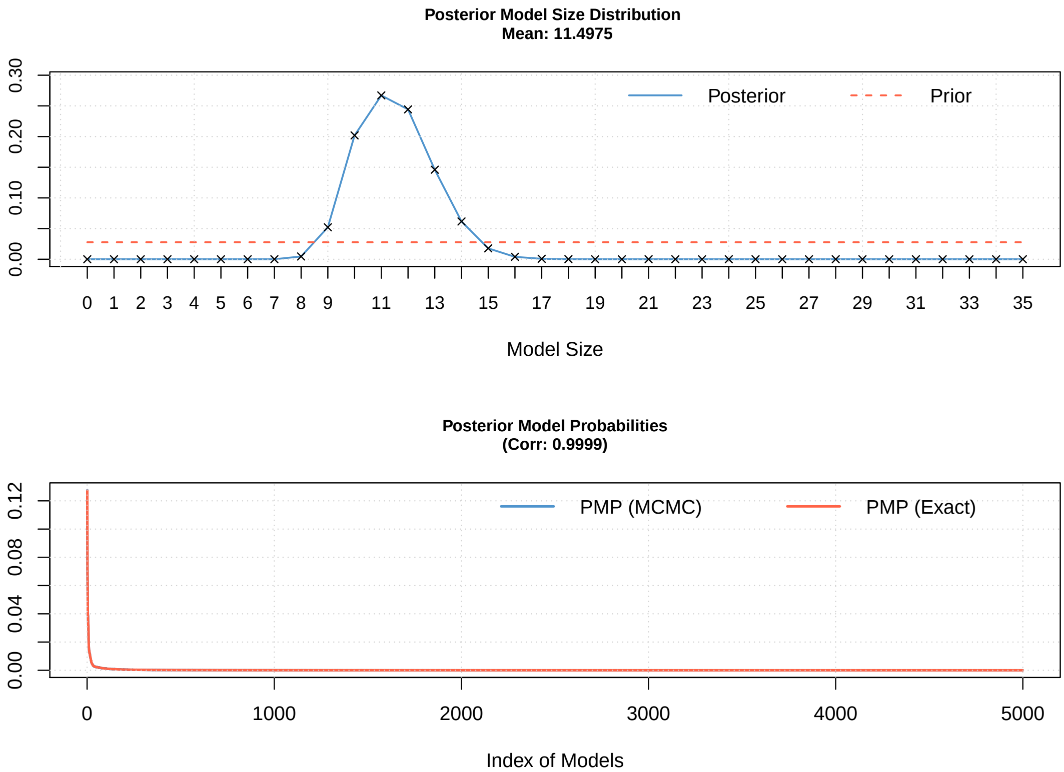
Notes: The figure depicts the posterior model size distribution and the posterior model probabilities of the trimmed-data BMA exercise using unit information prior and dilution prior.

Table 5: *Diagnostics of the BMA estimation (country variables)*

Mean no. regressors	*Draws*	*Burn-ins*	*Time*	*No. models visited*
11.4975	$5 \cdot 10^6$	$1 \cdot 10^6$	4.995831 mins	597,272
Modelspace	*Visited*	*Topmodels*	*Corr PMP*	*No. obs.*
$3.4 \cdot 10^1 0$	0.0017%	97%	0.9999	2,989
Model prior	*g-prior*	*Shrinkage-stats*		
Random-dilution / 22	UIP	Av = 0.9997		

Notes: In this specification we employ the priors suggested by Eicher *et al.* (2011), who recommend using the unit information prior (the prior has the same weight as one observation of data) and using the dilution prior suggested by George (2010), which accounts for collinearity.

Figure 8: *Model size and convergence of the BMA estimation (country variables)*



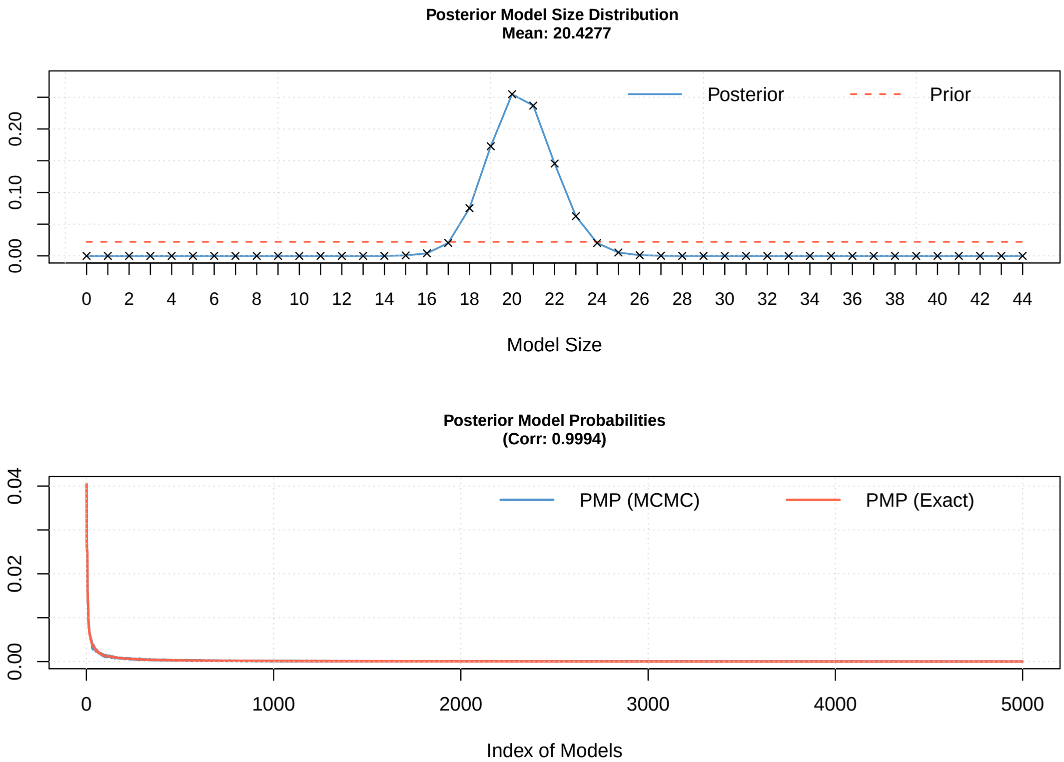
Notes: The figure depicts the posterior model size distribution and the posterior model probabilities of the cross-country BMA exercise employing the unit information prior and dilution prior.

Table 6: *Diagnostics of the BMA estimation (no survey data)*

Mean no. regressors	*Draws*	*Burn-ins*	*Time*	*No. models visited*
20.4277	$5 \cdot 10^6$	$1 \cdot 10^6$	6.198806 mins	727,647
Modelspace	*Visited*	*Topmodels*	*Corr PMP*	*No. obs.*
$1.8 \cdot 10^{1}3$	0.0000041%	91%	0.9994	2,943
Model prior	*g-prior*	*Shrinkage-stats*		
Random-dilution / 22	UIP	Av = 0.9997		

Notes: In this specification reported we employ the priors suggested by Eicher *et al.* (2011), who recommend using the unit information prior (the prior has the same weight as one observation of data) and using the dilution prior suggested by George (2010), which accounts for collinearity.

Figure 9: *Model size and convergence the BMA estimation (no survey data)*



Notes: The figure depicts the posterior model size distribution and the posterior model probabilities of BMA exercise using data WO survey datasets employing the unit information prior and dilution prior.

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